

Title

Nationwide telemetry-based cardiac arrhythmia screening in older adults: findings from a large pragmatic trial

Abstract

Background: Systematic identification of previously undiscovered atrial fibrillation in at-risk populations remains challenging despite its implications for stroke prevention.

Methods: We enrolled 50,549 adults aged 65 and older with at least one clinical marker for thromboembolism risk through a fully automated digital platform. Participants were randomly assigned to receive remote ECG telemetry surveillance or standard clinical care.

The telemetry group underwent 7 days of continuous electrocardiographic monitoring using a wearable patch system mailed directly to their homes. **Results:** Among participants allocated to telemetry, 1.9% were found to have previously unidentified atrial fibrillation. Of these newly detected cases, approximately 96% of patients initiated appropriate anticoagulation management following recommendations to their primary care providers. No serious adverse events related to the screening process were reported.

Conclusion: Large-scale, remotely administered ECG screening is operationally feasible and identifies clinically relevant arrhythmia in a significant proportion of older individuals at baseline thrombotic risk, with high uptake of guideline-directed medical therapy.